Clustering Claims for Loss Reserving: What Segmentation Can and Cannot Do

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Abstract Unsupervised clustering is often proposed as a data-driven way to segment claims before reserving. We test that proposal on the simulated SPLICE portfolios: the full-complexity scenario in its standard and covariate versions, and then all five SPLICE scenarios.

An ex-post clustering of completed claims finds one natural split, between small, quickly settled claims and the rest. K-means recovers it far more stably than Ward's hierarchical method (bootstrap adjusted Rand index 0.96 against 0.49). Finer partitions depend mainly on how correlated measures of claim complexity are weighted.

We then use only information available at the valuation date and backtest against the actual run-off. Segmenting a paid chain ladder with clusters never beats random segmentations of the same sizes, in any of the five scenarios. Adjusting the case estimates of open claims with factors fitted out of time is the most reliable approach: it cuts the RBNS error from about 40% to 7–12% with a single factor, and groups add little to it.

Groups do help in three narrower roles:

- identifying large claims to reserve separately, which is the best approach under an inflation shock;
- refining a model that accounts for calendar-period trends;
- monitoring whether the claim mix or the development speed is changing.

The analysis also shows how inflation drift and the timing of information can create apparent gains. We propose five evaluation criteria and a random-segmentation placebo for any clustering used in reserving.

Keywords: loss reserving; clustering; segmentation; K-means; hierarchical clustering; chain ladder; case estimates; SPLICE

1. Introduction

Loss reserving is, at its core, an exercise in managing heterogeneity. Claims differ in size, in the delay before they are reported, in the time they take to settle, and in the pattern of payments and case-estimate revisions they generate. Traditional methods deal with this by aggregating claims into run-off triangles, segmented in advance by line of business, type of cover or accident period, and by assuming that development is homogeneous within each segment (England & Verrall, 2002; Taylor, 2000). The segments are chosen by judgement, and the plausibility of the homogeneity assumption depends on how well that judgement was made.

Recent research has turned to individual-claim data and machine-learning methods (Wüthrich, 2018), supported by simulators that generate claim histories whose properties are fully known (Avanzi et al., 2021, 2023). Within this landscape, unsupervised clustering is an appealing tool.

Rather than imposing segments, it lets the data suggest groups of claims, or of portfolio cohorts, with similar development characteristics. Such groups could support segmentation *before* reserving, or help to interpret the results of a reserving analysis. This is the question we investigate: can clustering claims by their development characteristics yield segments that are useful to a reserving actuary?

Clustering is, however, unforgiving in a specific way. It has no target variable, so there is no held-out error to tell us whether the result is right. K-means, hierarchical and model-based algorithms will return a partition of any data set, and that partition depends on decisions taken before the algorithm is run: what counts as an object (a claim or an occurrence period), which variables describe it, on what scale, under which distance and into how many groups (Hastie et al., 2009; Jain, 2010). Each of those decisions can be defended only with knowledge of the data. An analyst who skips the data-understanding stage does not avoid making them; the analyst makes them implicitly, and without evidence.

This paper therefore treats understanding the data as the first stage of the clustering study, not as a preliminary to it. Using the SPLICE Complexity 5 portfolio (Avanzi et al., 2023), we validate the structure of the data, examine the distribution of the candidate features and the dependence between them, and choose the unit of analysis. Our aim is twofold: to report the features of the data that matter for reserving, and to show that each of them translates into a concrete modelling decision, from the transformation of variables to the choice of algorithm. We then use those decisions to test how clustering can be used in reserving: to segment a chain-ladder triangle, to adjust the case estimates of open claims, to set large claims apart, to refine a model with calendar-period trends, and to monitor the portfolio. All tests use only information available at the valuation date, are scored against the actual run-off, and are repeated on all five SPLICE scenarios.

Section 2 sets out the theoretical background. Section 3 presents the data and the exploratory analysis. Section 4 describes the clustering and reserving experiments, and Section 5 reports their results. Section 6 discusses the implications and limitations.

2. Theoretical Background

This section links three bodies of ideas that the rest of the paper relies on: segmentation as it is practised in reserving, clustering as a data-driven way of producing segments, and the timing of information, which decides whether a segment can be used at a valuation date at all. It closes with the criteria by which we judge whether a clustering is useful to a reserving actuary.

2.1 Reserving as a segmentation problem

Aggregate reserving methods project a triangle of cumulative amounts $C_{r,j}$, where r indexes the cohort (accident, occurrence or notification period) and j the development period. The chain ladder estimates development factors

$$\hat{f}_j = \frac{\sum_r C_{r,j+1}}{\sum_r C_{r,j}}, \tag{1}$$

the sums running over the cohorts observed at both ages, and projects each cohort from its latest diagonal. The method is distribution-free, but it is not assumption-free: it treats every cohort as following the same development pattern, so that the ratio of successive cumulative amounts does not depend on the cohort (Mack, 1993; Taylor, 2000).

That assumption is plausible only if the claims within the triangle are homogeneous, or if their mix is stable across cohorts. When a portfolio mixes, say, small claims that settle within a few quarters with large claims that develop for years, the aggregate factors are an average of two patterns weighted by the mix. A cohort whose mix differs from the historical average is then projected with the wrong pattern. Actuaries therefore segment before projecting, by line of business, type of cover, claim type or size band, and accept that a segment should be large enough to give credible factors (Friedland, 2010). Segmentation is a trade-off: finer segments are more homogeneous but thinner, and thin triangles give volatile factors, particularly in the later development periods, where few cohorts are observed.

Segmentation addresses heterogeneity between claims; it does not address changes over calendar time. If settlement speeds up, or if inflation accelerates, every segment's pattern shifts at once, and no partition of claims removes the resulting bias in (1). This distinction matters for evaluation: an improvement from segmenting should be judged against errors that segmentation can in principle remove.

Individual-claim reserving models avoid the triangle and model each claim's development directly (Antonio & Plat, 2014; Wüthrich, 2018). Clustering sits between the two traditions. It uses individual claim information, but its output, a set of groups, can feed an aggregate method such as the chain ladder, or can define the groups within which simpler claim-level adjustments are estimated.

2.2 Clustering as data-driven segmentation

Clustering partitions n objects described by feature vectors $x_1, \ldots, x_n \in \mathbb{R}^p$ into groups whose members are more similar to each other than to members of other groups. We use three families of method (Hastie et al., 2009; Jain, 2010).

K-means seeks K centroids $\mu_1, \ldots, \mu_K$ and an assignment $c(i)$ that minimise the within-cluster sum of squares

$$W = \sum_{k=1}^{K} \sum_{i:\, c(i)=k} \|x_i - \mu_k\|^2, \tag{2}$$

usually by alternating assignment and centroid updates from several random starts (Lloyd, 1982; MacQueen, 1967). Because W is a sum of squared Euclidean distances, K-means favours compact groups of similar spread and is sensitive to the scale of the features and to extreme observations.

Agglomerative hierarchical clustering starts from singletons and merges, at each step, the pair of clusters that is closest under a linkage rule. With Ward's rule each merge is the one that increases W the least (Ward, 1963), so Ward and K-means optimise the same criterion, one greedily and one iteratively. The sequence of merges, displayed as a dendrogram, shows the partitions for every number of clusters at once, and large jumps in merge height suggest natural cut points. Because merges are never undone, an early merge that depends on a few observations is carried into every coarser partition.

Model-based clustering assumes that the data come from a finite mixture of distributions, typically Gaussian, and assigns each object to the component with the highest posterior probability (McLachlan & Peel, 2000). It allows clusters of different shapes and sizes, and the number of components can be chosen by an information criterion. This makes it a natural cross-check for groups that are not compact, such as the band of quickly settled small claims found in Section 3.5.

All three methods require the analyst to fix the features, their transformation and scale, the distance, and the number of clusters. None of these can be learned from a fit criterion, because a partition into more groups always reduces W. Internal indices that trade compactness against separation help to compare candidate numbers of clusters: the silhouette, which compares each object's average distance to its own cluster with that to the nearest other cluster (Rousseeuw, 1987); the Calinski–Harabasz index, a ratio of between- to within-cluster dispersion (Caliński & Harabasz, 1974); and the Davies–Bouldin index, which penalises clusters that are wide relative to their separation (Davies & Bouldin, 1979). These indices measure geometry, not usefulness. A clear geometric split need not matter for reserving, and a useful segmentation need not be geometrically well separated.

Practical implication. For a reserving actuary, a cluster is a proposed segment. Like a segment chosen by judgement, it has to be defensible: we must be able to say which features define it, why they are weighted as they are, and how sensitive the grouping is to those choices.

2.3 Information available at the valuation date

Reserving is done at a valuation date τ, and only information available at τ can be used. At τ each claim is in one of three states: not yet reported (IBNR), reported and open (RBNS), or settled. For a reported claim the information available consists of its characteristics at notification and its transaction history up to τ: payments, case-estimate revisions and their timing. Its ultimate cost, final settlement delay and final number of payments are outcomes that the reserve is meant to estimate, not inputs to it.

A feature is admissible at τ if it can be computed from the data truncated at τ. Admissibility can be checked mechanically: deleting every transaction after τ must leave the feature unchanged. Features that fail this test introduce leakage, in the sense of Kaufman et al. (2012). The leak can be indirect: a "proportion paid" computed with the ultimate cost as its denominator uses future information even though its numerator does not. This yields two distinct exercises. An *ex-post* clustering of settled claims on their final outcomes describes completed claim profiles; it can motivate a segmentation but cannot assign an open claim to a segment. An *ex-ante* clustering on admissible features can be used at τ.

Admissibility is necessary but not sufficient when the clusters are to segment a triangle. A label computed from a claim's history up to τ means different things for a claim reported last quarter and one reported thirty quarters ago, because the two have been observed for different lengths of time. Grouping cohorts by such a label mixes development ages into the definition of the segment and distorts the per-segment factors. A label used for triangle segmentation should therefore be

measured at a fixed development age, for example at notification, so that it means the same thing in every cohort. Labels built on the full history at τ remain useful, but for a different purpose: describing the open claims at τ, for instance to adjust their case estimates.

Two further consequences follow. First, monetary features measured in nominal terms drift with inflation, so a fixed partition of nominal amounts moves later cohorts into higher groups. Such features need to be expressed relative to a contemporaneous reference before clustering. Second, the reserve for reported claims and the reserve for claims not yet reported are different problems. Clusters of reported claims say nothing directly about claims that have not yet been notified, so we evaluate them on the reported claims only, using triangles organised by notification period.

2.4 Criteria for a useful clustering

Without a target variable, the quality of a clustering has to be judged against its purpose. For reserving we assess the following five criteria.

i. *Interpretable, distinct development.* Clusters should differ in ways an actuary can describe, in size, speed or payment pattern, and their development patterns should differ enough for separate projection to matter. Similar patterns imply that segmentation can change little.

ii. *Assignability at the valuation date.* Every feature used to define the clusters must be admissible at τ, and the assignment rule fitted on past data must be applicable unchanged to new claims.

iii. *Stability.* Clusters should not depend on the particular sample or period. We measure agreement between partitions with the adjusted Rand index (Hubert & Arabie, 1985) and assess stability by refitting on bootstrap samples (Hennig, 2007). Over time, a stable segmentation keeps a broadly constant mix across cohorts unless the portfolio itself changes.

iv. *Credibility.* Each cluster must contain enough claims and development history to support estimation. In a triangle this means enough cohorts observed at each development age.

v. *Predictive usefulness.* The decisive test is whether the clusters improve estimates of future payments relative to an agreed baseline, assessed out of time: fit at one valuation date and score at a later one against the actual run-off. Two baselines are needed. An unsegmented method shows whether segmentation helps at all. A random partition with the same group sizes acts as a placebo, separating the effect of the clusters' content from the mechanical effect of splitting the data. Simple rule-based segmentations, such as bands of a single variable, show whether the clustering adds anything that an actuary's rule would not.

The criteria can conflict. More clusters improve homogeneity and may improve interpretability, but reduce credibility and usually stability. We report all five, rather than a single index, so that the trade-off is visible.

3. Data and Exploratory Analysis

Before any clustering algorithm is run, the analyst must decide what is being clustered, on what scale and with which variables. This section documents those decisions for the SPLICE Complexity 5 portfolio (original analysis: `notebooks/01_data_exploration.ipynb`; figures and supplementary checks: `article/make_figures.py`).

3.1 Why understanding the data comes first

Clustering is often described as letting the data speak. In practice the data speak only through the analyst's preparation of them, and four considerations explain why that preparation has to be informed.

First, clustering algorithms always return an answer. K-means, hierarchical and model-based methods will partition any data set into the requested number of groups, whether or not it contains natural groups, and without a target variable there is no held-out error to say whether the result is meaningful (Jain, 2010). Second, distance-based methods are sensitive to the scale, skewness and redundancy of their inputs. A variable spanning six orders of magnitude swamps one that counts payments, a few extreme observations can pull centroids towards them, and several variables that measure the same phenomenon give it several votes; even the choice of standardisation can materially change how well structure is recovered (Hastie et al., 2009; Milligan & Cooper, 1988). Third, the unit of analysis and the timing of information determine what a cluster *means*. A cluster of completed claims describes history, whereas a cluster of open claims at a valuation date can support prediction, and mixing the two is a form of information leakage (Kaufman et al., 2012). Fourth, data errors do not stay local: a broken link between tables or a misread missing value is absorbed silently into the features, so the clusters reflect the plumbing of the data rather than the behaviour of claims.

Practical implication. Reserving practice already rests on reconciliation and on knowing what is available at the valuation date; what changes is that, with nothing to fit, no residual warns us that a preparation choice was wrong. Exploratory work is therefore part of the model (Tukey, 1977), since summary statistics alone can conceal the structure that matters (Anscombe, 1973).

3.2 The SPLICE Complexity 5 portfolio

We use SPLICE, a simulator of individual claim histories that models payments and the evolution of case estimates (Avanzi et al., 2023), built on the SynthETIC claim simulator (Avanzi et al., 2021). Simulated data suit a first study because they are complete and because the mechanisms that generate heterogeneity are known, which will later let us check whether clusters recover something real. The price is realism: conclusions will need confirming on real data.

We take the Complexity 5 scenario (random seed 42) from the example data distributed with SPLICE.[1] It is the most complex of the five published scenarios, with dependence between claim size and both delays, and between settlement delay and occurrence period. The portfolio has 3,663 claims from 40 quarterly occurrence periods, held in three tables linked by the claim identifier: claims (one row per claim), payments (19,322 rows) and incurred transactions (31,554 rows, recording case-estimate revisions and cumulative payments). Times are in quarters; amounts are in unspecified currency units.

[1] https://github.com/agi-lab/SPLICE, folder `datasets/complexity_5`.

3.3 Structural validation

We validated the tables before examining any distribution, leaving the raw values unmodified. The claim identifier is unique, all 3,663 claims appear in both transaction tables, no transaction refers to an unknown claim, and the number of payment records per claim agrees with the recorded number of payments in every case. The only missing entries are 9,593 revision multipliers in the incurred table, which are structural rather than defects: they coincide exactly with the payment-only transactions, in which no case-estimate revision occurs. The nominal payments of each claim also sum to its recorded claim size to numerical precision. The tables therefore form a coherent relational data set, and no imputation or repair is needed.

3.4 Marginal distributions

Our baseline description of a claim uses four variables from the claims table: claim size (ultimate severity), notification delay, settlement delay and the number of payments, which capture how large a claim is, how quickly it is reported, how long it takes to settle and how many payments it involves (Table 1, Figure 1).

Table 1: Descriptive statistics of the candidate clustering variables ($n = 3{,}663$), with skewness on the raw scale and after a $\log(1+x)$ transformation.

					Skewness	
Variable	Mean	Median	95th pct.	Maximum	Raw	Log
Claim size	166,924	77,185	618,724	3,130,790	3.69	−1.01
Notification delay	2.14	1.74	5.37	12.13	1.36	0.17
Settlement delay	9.80	7.55	27.23	56.29	1.37	−0.31
Number of payments	5.27	4	11	30	2.13	−0.03

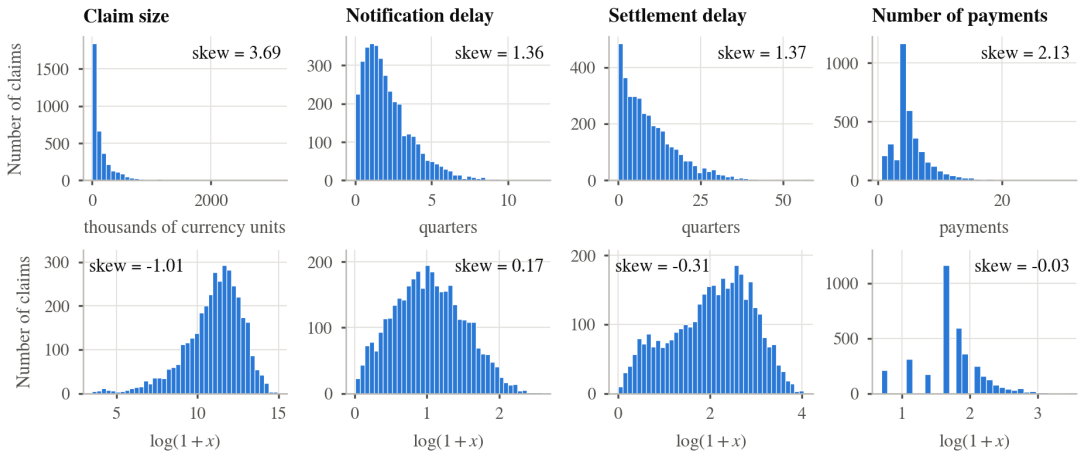


Figure 1: Marginal distributions of the four candidate variables on the raw scale (top) and after a $\log(1+x)$ transformation (bottom), with sample skewness.

All four variables are right-skewed and measured on very different scales. Claim size is the most extreme: its mean is more than twice its median, the largest claim is about 41 times the median claim, and the largest 10% of claims account for 46% of total claim cost. Euclidean distance on the raw variables would let claim size decide almost every distance. A $\log(1+x)$ transformation reduces the skewness of every variable to about one or less in absolute value, and standardisation then puts them on a comparable footing.

The logarithm is not a cure-all, however. For claim size it slightly over-corrects, leaving a left tail of very small claims (116 below 1,000, of which 27 below 100). Log settlement delay has a shoulder at short delays, hinting at a group of quickly settled claims (Section 3.5). The number of payments is discrete, with a sharp mode at four payments (1,166 claims, about a third), so no transformation makes it look continuous. The extreme claims are valid simulated observations and we retain them: in reserving, large and long-tailed claims are often exactly the segment of interest. *Practical implication:* transform and standardise before computing any distance, but do not delete the tail; assess its influence by sensitivity analysis instead.

3.5 Dependence between candidate features

Because the variables are skewed and one is discrete, we measure dependence with Spearman rank correlation (Figure 2a). Claim size is strongly associated with both settlement delay ($\rho = 0.68$) and the number of payments ($\rho = 0.68$), and settlement delay with the number of payments ($\rho = 0.54$). Notification delay is weakly and negatively related to the others (ρ between -0.29 and -0.19). The Pearson correlations for the first three pairs are markedly lower (0.45, 0.39, 0.37), so the relationships are monotone but far from linear: larger claims stay open longer and involve more payments.

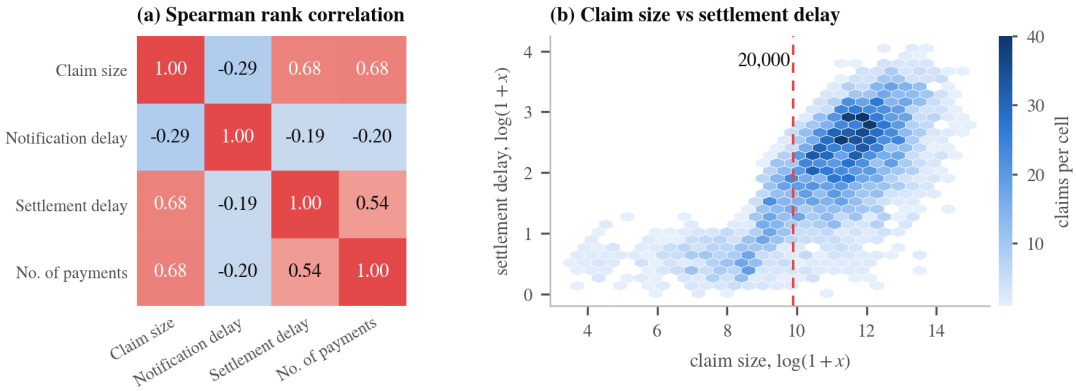


Figure 2: (a) Spearman rank correlations between the candidate variables. (b) Claim size against settlement delay on the $\log(1+x)$ scale; the dashed line marks a claim size of 20,000.

Claim size, settlement delay and the number of payments are therefore largely three measurements of one dimension, which one might call claim complexity. Under equal-weight distances that dimension is counted three times, and clusters are likely to be organised around it. We keep the four variables at first but plan a sensitivity analysis over alternative feature subsets.

Figure 2b shows a second feature that the correlation matrix hides: the joint distribution is not a single elliptical cloud. Claims below roughly 20,000 (833 claims, 23% of the portfolio) settle far more quickly than the rest, with a median settlement delay of 1.4 quarters against 10.2, and form a distinct band of short delays rather than the lower end of the main cloud. This is consistent with the scenario's specification (mean settlement delay grows with the logarithm of claim size) and explains the shoulder in Figure 1. It matters because K-means favours compact, roughly spherical groups (Hastie et al., 2009), so shapes like this argue for also using hierarchical and model-based clustering.

3.6 Choosing the unit of analysis

A reserving analyst can cluster individual claims or aggregated portfolio cohorts, and the choice changes the question being asked. The cohort alternative is to summarise each occurrence period

by its claim frequency and typical claim characteristics and to look for periods that behave alike. We built this period-level table (40 rows) and examined it for regimes (Figure 3). Claim frequency ranges from 73 to 111 claims per period (mean 91.6), and the median claim size, median settlement delay and mean number of payments fluctuate around stable levels without persistent breaks. The only visible drift is a mild fall in settlement delay across periods (Spearman correlation of -0.06 with occurrence period), consistent with the scenario's design. Period-level clustering would also cut the sample from 3,663 observations to 40 and would conceal the claim-level heterogeneity seen above: the share of claims below 20,000 is essentially constant over time (22% in periods 1–20, 23% in periods 21–40), so period medians barely register the quick-settling group.

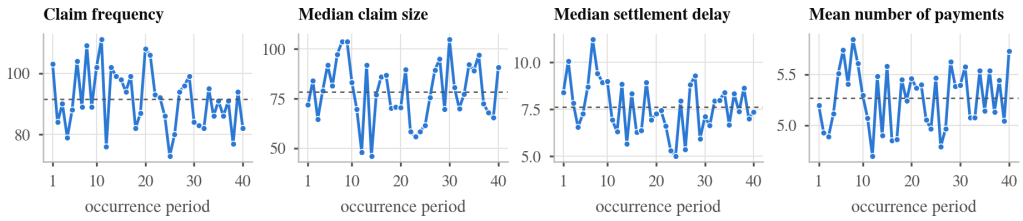


Figure 3: Occurrence-period summaries (40 periods); the dashed line is the average over periods. Claim size is in thousands of currency units and settlement delay in quarters.

We therefore take the individual claim as the primary unit of analysis, since it gives enough observations and preserves variation between claims; occurrence-period summaries remain a secondary diagnostic for the resulting clusters.

3.7 An ex-post baseline

The baseline variables include ultimate claim size, final settlement delay and final number of payments, none of which is known until a claim has settled. Our first clustering exercise is consequently an *ex-post segmentation*: it can describe different types of completed claim, but it cannot assign a claim to a cluster at notification or at a valuation date. A reserving application must separate features available at the valuation date from the outcomes the analysis is meant to understand or predict (Kaufman et al., 2012). Aggregating the payment and incurred tables to one row per claim can supply such features, for example time to first payment, proportion paid at given development ages, and the number and size of incurred revisions. The covariate-adjusted portfolio (files with suffix `_cov`) is a separate simulation, kept for robustness checks.

3.8 From findings to design decisions

Table 2 traces each finding to a design decision. None is exotic, but each is invisible to the algorithm and consequential for its output, so documenting them lets the results that follow be interpreted and challenged.

Table 2: Findings of the data-understanding phase and the design decisions they imply.

Finding	Design decision
All four variables are right-skewed, on very different scales, and extreme claims are valid.	Apply $\log(1+x)$ and standardise before computing distances; retain extreme claims and test their influence.
Claim size, settlement delay and payment count are strongly correlated (ρ 0.54–0.68).	Test alternative feature subsets and weightings.
Small claims form a distinct, quickly settled band (non-elliptical).	Compare K-means with hierarchical and model-based clustering.
No persistent regimes across occurrence periods.	Cluster claims; use period as a secondary diagnostic.
Baseline variables are known only at settlement.	Label the baseline ex-post; derive valuation-date features from transaction tables.

4. Methodology

The study has two parts. The first is an *ex-post* clustering of completed claims on the baseline variables of Section 3. It asks which groups the data contain, and how much the answer depends on the algorithm and on the analyst's choices. The second is a set of *valuation-date* experiments. They ask whether groups built only from information available at the valuation date improve reserve estimates. Code for every step is in the project repository (`src/`, `scripts/`, notebooks 02 and 03).

4.1 Data versions and monetary conventions

We use two portfolios: the standard Complexity 5 portfolio of Section 3, and a covariate-adjusted portfolio in which claim size depends on three claim characteristics: legal representation, injury severity and claimant age. The covariate portfolio is the one published with SPLICE. The published files do not include the covariate assigned to each claim, so we recover the labels by regenerating the portfolio with the released package versions, which reproduce the published files exactly (Appendix A). The two portfolios share occurrence times, so both have 3,663 claims, but they differ in everything drawn after the covariates are assigned and are not mixed.

The claim size in the claims table is expressed in constant money. The cash actually paid includes base and superimposed inflation. It is recorded per payment and accumulates to the cumulative paid amount in the incurred table. In Complexity 5 total inflated payments are 1.83 times total constant-money claim size; the ratio averages 3.7 per claim, because small claims carry the most superimposed inflation. All reserving quantities below therefore use inflated payments.

4.2 Ex-post clustering

The objects are the 3,663 claims, described by claim size, notification delay, settlement delay and number of payments. Following the decisions in Table 2, each variable is transformed by $\log(1+x)$ and standardised, and Euclidean distance is used.

We fit K-means (20 random starts) and agglomerative clustering with Ward linkage for $k = 2, \ldots, 8$, and choose k with four tools:

- the Ward dendrogram;
- the silhouette;
- the Calinski–Harabasz index;
- the Davies–Bouldin index.

We compare the two algorithms' partitions with the adjusted Rand index (ARI).

Stability is measured by refitting each method on 20 bootstrap samples and computing the ARI between the bootstrap labels and the full-data labels of the drawn claims. Comparing on the drawn claims avoids having to extend a hierarchical solution to unseen points.

Sensitivity to correlated features is tested by refitting K-means on alternative feature sets and measuring the ARI against the baseline partition:

- dropping the number of payments;
- dropping settlement delay;
- claim size and notification delay only;
- a single complexity index (the first principal component of size, settlement delay and number of payments) with notification delay;
- all four features decorrelated by principal-component whitening.

Sensitivity to extreme claims is tested by removing the largest 1% of claims.

4.3 Valuation-date features

For a valuation date τ we keep the claims reported by τ and cut every transaction history at τ.

Features at notification are used to segment triangles (Section 2.3):

- the notification delay;
- the initial case estimate set at notification;
- in the covariate portfolio, the three covariates.

The nominal initial case estimate drifts with inflation: its median rises from about 40,000 in notification periods 1–10 to 88,000 in periods 31–40. We therefore divide it by the median initial estimate of claims notified in the same period. This quantity is available at τ because a period's claims are all notified by its end.

Features at the valuation date describe the open claims at τ. They follow the list proposed in our working-party discussions:

- notification delay and age since notification;
- cumulative paid and number of payments to date;
- time since the most recent payment;
- the current case estimate;
- the number of case-estimate revisions since notification and the largest absolute log revision.

Monetary features enter in scale-free form, relative to the claim's own initial case estimate or, for the initial estimate itself, relative to its notification period.

No feature uses the ultimate claim size, the final settlement delay or the final number of payments. Admissibility is checked by an automated test: deleting every transaction after τ leaves all features unchanged.

4.4 Experiment 1: segmenting the chain ladder

At $\tau \in \{20,24,28,32,36,40\}$ we build triangles of inflated cumulative paid amounts with rows indexed by *notification* period. A notification cohort contains only reported claims, so the chain ladder estimates the RBNS reserve. That estimate can be compared directly with the actual payments after τ on the same claims. IBNR is excluded.

Factors follow (1) without a tail. A factor estimated from fewer than five cohorts is set to one. Without this rule, a single late payment in the oldest cohort gave $\hat{f}_{34} = 1.43$ at $\tau = 36$ and raised the whole reserve by about 40%. The rule is applied identically to every segment and method.

A segmented estimate is the sum of chain-ladder reserves computed separately on each segment's triangle. Clusters are refitted at each τ on the claims reported by then, using notification features, with K-means or Ward and $k = 2,\ldots,5$. We compare them with:

- the unsegmented chain ladder;
- the outstanding case estimates;
- equal-count bands of the relative initial case estimate, a simple actuarial rule;
- random groups with the same sizes as the K-means clusters, drawn 30 times, as a placebo for the mechanical effect of splitting the data;
- in the covariate portfolio, segmentation by the true injury-severity groups $\{1,6\}$, $\{2\}$ and $\{3,4,5\}$.

The main metric is the total error, $100\,(\hat{R}-R)/R$, where R is the actual RBNS run-off. We also record the sum of absolute row errors.

4.5 Experiment 2: case-estimate factors for open claims, out of time

The second experiment uses the clusters for a different purpose: to adjust the case estimates of open claims. At a fitting date τ_0 the open claims are grouped. For each group g, a development

factor is computed from data available at the later application date $\tau_1 = \tau_0 + 8$:

$$\hat{\lambda}_g = \frac{\sum_{i \in g} \left[P_i(\tau_0, \tau_1] + O_i(\tau_1) \right]}{\sum_{i \in g} O_i(\tau_0)}, \tag{3}$$

where $P_i(\tau_0, \tau_1]$ is claim i's payments in the window and $O_i(t)$ its outstanding case estimate at t.

At τ_1 the open claims are assigned to groups with the rule fitted at τ_0 (the same scaler and centroids, or the same band edges). The RBNS reserve is then estimated as $\sum_g \hat{\lambda}_g \sum_{i \in g} O_i(\tau_1)$ and scored against the actual payments after τ_1. A second automated test checks that deleting all data after τ_1 leaves the estimate unchanged.

The numerator of (3) still contains case estimates at τ_1 for claims that remain open. When case estimates are short, the factor therefore understates full development. The bias is common to every grouping, so comparisons between groupings remain like for like.

We compare the following groupings, for $\tau_1 \in \{24, \ldots, 40\}$:

- a single portfolio-wide factor;
- quartile bands of claim age;
- quartile bands of the ratio of current to initial case estimate;
- K-means clusters on the valuation-date features with $k = 3, 4, 6$;
- K-means with the covariates added, in the covariate portfolio.

Raw case estimates and the unsegmented paid chain ladder at τ_1 are reported as reference points.

4.6 Experiment 3: other uses of groups, across the five scenarios

Experiments 1 and 2 use the groups to split a triangle or to adjust case estimates. Experiment 3 tests three further uses. It then repeats all of them, together with the main variants of Experiments 1 and 2, on the standard portfolios of the five SPLICE scenarios:

- Complexity 1: simple, homogeneous claims, no inflation;
- Complexity 2: size-dependent delays and 2% annual inflation;
- Complexity 3: settlement that speeds up over time;
- Complexity 4: an inflation shock at time 30;
- Complexity 5: the full-complexity scenario used above.

The valuation dates are $\tau \in \{24, 28, 32, 36, 40\}$, so that every method, including those fitted eight quarters earlier, can be evaluated at every date. Groups are K-means clusters with $k = 3$ on the notification features of Section 4.3, refitted at each τ. The cluster with the largest median relative initial estimate is called the *large* cluster.

Monitoring. At $\tau = 40$ we track, for each notification cohort and cluster, the cluster's share of the cohort and the share of its claims settled within eight quarters of notification. The second quantity is observable for cohorts notified up to period 32. A trend is summarised by the Spearman correlation with the notification period, and its p-value. The monitor is meant to distinguish a change in the mix of claims from a change in development speed that affects every group, which calls for a different reserving response (Section 2.1).

Large claims apart. The claims of the large cluster are removed from the triangle. The remaining claims are projected with the chain ladder of Section 4.4. The open large claims are reserved as their case estimates times a factor (3), fitted at $\tau - 8$ on the open large claims of that date. As a rule-based alternative, the large group is defined instead as the same number of claims with the highest relative initial case estimate.

Groups inside a model with calendar effects. Let $X_{r,j,g}$ be the incremental inflated payments in development period j of claims notified in period r and belonging to group g, and $n_{r,g}$ the number of such claims. We fit a quasi-Poisson generalised linear model

$$\log \mathrm{E}[X_{r,j,g}] = \log n_{r,g} + a_{j,g} + \gamma \frac{r+j}{\tau} + \beta \frac{r}{\tau} \log(1+j), \tag{4}$$

where $a_{j,g}$ is a development pattern, γ a linear calendar-period trend, and β a trend in the speed of development across cohorts. We consider four versions:

- without the calendar trend ($\gamma = \beta = 0$);
- with the calendar trend only ($\beta = 0$);
- with both trends;
- with the trends and group-specific development patterns $a_{j,g}$, against a single pattern a_j.

The trends are common to all groups. As with the chain ladder, only development periods observed in at least five cohorts are fitted and projected. The future cells are projected with the fitted trends extrapolated, and the reserve is their sum.

A third automated test checks that the large-claims and GLM estimates are unchanged when all data after τ are deleted.

5. Results

5.1 Ex-post structure: one natural split

The data contain one clear division. In the Ward dendrogram the merge that joins the last two clusters is far higher than any other: 105.9, against 60.7 for the next merge and 42.0 for the one after. All three internal indices prefer $k = 2$ for both algorithms (Table 3, Figure 4a). Beyond two clusters the indices flatten, with no second natural number of groups.

The two-cluster solution separates 932 small, quickly settled claims from the remaining 2,731:

- *small, fast claims:* median size 8,500, settled in a median of 1.5 quarters with 2 payments;
- *all other claims:* median size 125,000, settled in a median of 10.6 quarters with 5 payments.

Eighty-three per cent of the first group are below 20,000, so this is the short-delay band identified in Section 3.5. It accounts for only 1.9% of total claim cost.

Table 3: Ex-post clustering of the 3,663 claims: internal diagnostics and bootstrap stability. Silhouette and Calinski–Harabasz (CH): higher is better; Davies–Bouldin (DB): lower is better. Stability is the mean adjusted Rand index (ARI) over 20 bootstrap refits.

	K-means				Ward			
k	Silh.	CH	DB	Stability	Silh.	CH	DB	Stability
2	0.40	2,563	1.00	0.96	0.34	2,272	1.17	0.49
3	0.27	2,128	1.29	0.94	0.26	1,895	1.20	0.51
4	0.25	1,911	1.19	0.80	0.20	1,609	1.55	0.41
5	0.25	1,748	1.25	0.79	0.20	1,490	1.35	0.41

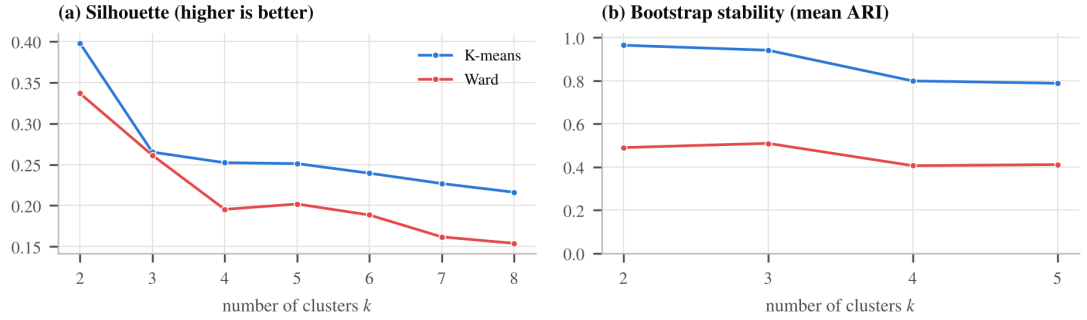


Figure 4: Ex-post clustering by number of clusters: (a) silhouette; (b) bootstrap stability (mean ARI between bootstrap and full-data partitions).

K-means is preferable to Ward on these data. It scores better on every index at almost every k. It is also far more stable: its bootstrap ARI is 0.96 at $k = 2$ and 0.94 at $k = 3$, against about 0.5 for Ward at every k. Ward's greedy merges are never undone, so a change in a few early merges propagates to the final partition. The two algorithms agree only moderately (ARI 0.59 at $k = 2$, 0.33–0.50 above).

Beyond two clusters, the partition reflects modelling choices more than structure. Removing one of the three correlated complexity variables changes the K-means partition substantially (ARI against the baseline 0.52–0.74 for $k = 2, 3, 4$). Replacing them by a single complexity index, or decorrelating all four features, changes it more (ARI 0.26–0.61). By contrast, removing the largest 1% of claims leaves the partition almost unchanged (ARI ≥ 0.97).

At $k = 4$ the groups are interpretable:

- very small, fast claims (581 claims, 0.5% of cost);

- large, long-tailed claims with many payments (1,180 claims, 69% of cost; median settlement delay 16 quarters, 7 payments);
- two mid-size groups that differ mainly in notification delay (median 3.3 against 1.0 quarters).

The mix of these groups is stable across occurrence periods. Which finer partition is "right", however, depends on how much weight complexity should receive, and that is an actuarial judgement rather than a property of the data.

Practical implication. Clustering finds the obvious split between small fast claims and the rest without supervision. Finer segments need an explicit justification for the feature weights, and should be reported together with their sensitivity.

5.2 Experiment 1: segmenting the chain ladder

At $\tau = 40$ the standard portfolio has:

- 3,462 reported claims, of which 821 are open, and 201 IBNR claims;
- 645 million paid to date;
- actual RBNS run-off of 386.5 million;
- outstanding case estimates of 252.5 million, 35% short.

The shortfall of the case estimates is remarkably steady, between 34% and 45% at every valuation date in both portfolios.

Table 4: Paid chain ladder RBNS reserve error by segmentation: mean absolute total error over six valuation dates ($\tau = 20, \ldots, 40$), and total error at $\tau = 40$. Clusters use features at notification; $k = 3$ unless stated.

	Mean $\lvert\text{error}\rvert$ (%)		Error at $\tau = 40$ (%)	
Method	Standard	Covariates	Standard	Covariates
Unsegmented chain ladder	31.8	21.2	35.0	28.3
K-means, notification features	33.6	22.7	34.0	30.8
K-means, notification features, $k = 4$	30.6	21.4	34.9	28.9
Ward, notification features	30.5	20.7	35.3	29.3
K-means, notification features + covariates	–	23.0	–	22.8
True injury-severity groups	–	20.6	–	22.7
Bands of relative initial case estimate	32.1	21.6	34.6	31.1
Random groups, mean of 30	32.2	21.4	35.5	29.2
5th–95th percentile at $\tau = 40$			33.9–37.1	27.1–31.6
Case estimates	37.4	38.2	−34.7	−37.4

Table 4 and Figure 5 show three things.

First, every method moves together over time. Every segmentation underestimates by 45–59% at $\tau = 20$ and overestimates from $\tau = 32$ onwards, by up to 80% at $\tau = 36$ in the standard portfolio.

- The early underestimates reflect the missing tail: with only 20 notification periods, late development is not yet observed.
- The later overestimates reflect two features of Complexity 5 that the chain ladder cannot accommodate: the steady acceleration of settlement over occurrence periods, and superimposed inflation.

Both are calendar-period effects, and no partition of claims removes them (Section 2.1).

Second, segmenting by clusters gives no robust gain. In both portfolios, clusterings on the notification features lie within a few points of the unsegmented chain ladder and of the random groups on average, and at $\tau = 40$ inside the range of the random groups. The best method changes with k and with the algorithm, which is what one expects of noise. The paid development patterns of the notification clusters are close to the aggregate pattern, so separate projection has little to correct. Banding the initial case estimate is no better.

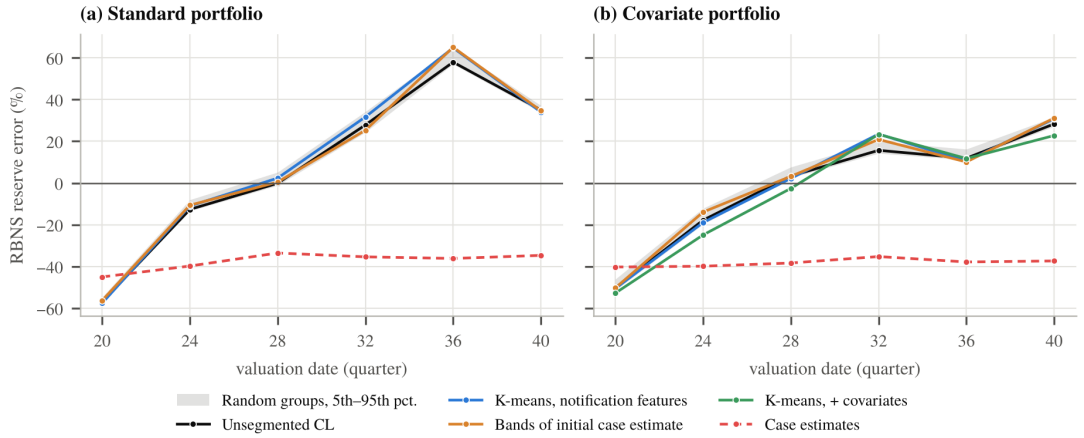


Figure 5: RBNS reserve error of the paid chain ladder by valuation date, unsegmented and segmented. The shaded band is the 5th–95th percentile of 30 random segmentations with the cluster sizes.

Third, the covariates help at some dates but not on average. At $\tau = 40$, K-means with covariates and the true severity groups give +22.8% and +22.7%, below the range of the random groups (+27.1% to +31.6%). Over all six dates, however, the severity groups are only 0.6 points better than the unsegmented chain ladder (20.6% against 21.2%), and K-means with covariates is worse (23.0%). A gain at a single valuation date is therefore not evidence of a useful segmentation, which is why we report the backtest over several dates.

Practical implication. When the dominant source of error is a calendar-period effect, segmenting the triangle by claim type is unlikely to help. Any claimed gain should be compared with random segmentations of the same sizes.

5.3 Experiment 2: adjusting case estimates

The open claims at $\tau = 40$ show strong differences in case-estimate adequacy. A K-means grouping into four clusters on the valuation-date features separates them mainly by claim age:

- *just reported, nothing paid* (264 claims, median age 1.9 quarters): actual future payments are 2.65 times the case estimate;
- *mid-age claims* (308 claims, median age 7.9 quarters): a ratio of 1.48;
- *the two older groups* (249 claims): the case estimate is about right, with ratios of 0.97 and 1.01.

In-sample, the clusters together with the case estimate explain 45% of the variance of log future payments, against 18% for the case estimate alone.

Whether that difference can be exploited is a question for the out-of-time test (Table 5, Figure 6).

Adjusting case estimates is far more effective than segmenting the triangle. A single portfolio-wide factor, fitted eight quarters earlier, reduces the error from 36–38% for raw case estimates to 7–8%. That is better than the paid chain ladder in both portfolios, and the reason is the stability of the case-estimate shortfall noted above.

Grouping adds little. In the standard portfolio, clusters with $k = 3$ or 4 reduce the mean error by 2.2–2.5 points relative to the single factor, but a simple rule does better: bands of the ratio of current to initial case estimate give 3.5%. In the covariate portfolio no grouping improves on the single factor (7.8–8.6% against 7.9%), with or without the covariates. The clusters therefore add nothing over a rule an actuary could set by hand.

Groups improve the total, not individual claims. The median absolute log error per claim is lowest for the single factor, or for six clusters in the standard portfolio, and no grouping improves it consistently.

Table 5: Out-of-time test: RBNS reserve error when factors (3) fitted at $\tau_1 - 8$ are applied at τ_1. Mean absolute total error over $\tau_1 = 24, \ldots, 40$, and median absolute log error per claim.

	Mean |error| (%)		Claim-level |log error|	
Method	Standard	Covariates	Standard	Covariates
Raw case estimates	35.9	37.7	–	–
Unsegmented paid chain ladder	26.7	15.4	–	–
Single factor	7.4	7.9	0.55	0.52
Bands of claim age ($k = 4$)	6.3	7.8	0.61	0.59
Bands of case estimate / initial estimate ($k = 4$)	3.5	8.6	0.64	0.64
K-means clusters ($k = 3$)	4.9	8.3	0.66	0.64
K-means clusters ($k = 4$)	5.2	8.0	0.68	0.63
K-means clusters ($k = 6$)	5.7	7.8	0.51	0.63
K-means clusters + covariates ($k = 4$)	–	8.2	–	0.63

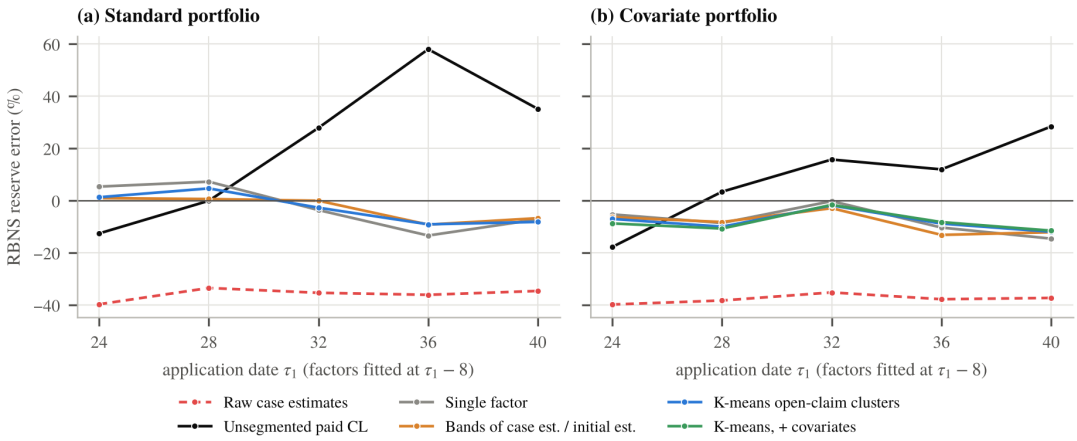


Figure 6: Out-of-time RBNS reserve error at the application date τ_1, for case-estimate factors fitted eight quarters earlier, compared with raw case estimates and the unsegmented paid chain ladder.

All groupings drift negative at the later dates, to between -6% and -13% at $\tau_1 = 36$ and between -2% and -15% at $\tau_1 = 40$. This is the understatement built into (3), and it affects them all.

5.4 Stability of the valuation-date clusters

Expressing the initial case estimate relative to its notification period removes most of the drift. The largest change in cluster share between notification periods 1–20 and 21–40 falls from 11.5 to 3.8 percentage points in the standard portfolio, and from 11.8 to 3.6 points in the covariate portfolio. Bootstrap stability of the notification clusters rises from 0.70 to 0.96 at $k = 3$ in the standard portfolio, and is 0.96 in the covariate portfolio.

Scale-free features matter for the open-claim clusters too. Clustering the open claims at $\tau = 40$ on nominal amounts produces a group of 181 claims with small case estimates whose actual future payments are 9.9 times their case estimate. The group disappears with scale-free features, where the ratios range from 0.97 to 2.65 (Section 5.3), so the extreme ratio depended on measuring amounts in nominal terms.

The groups that use the covariates are the least balanced. One open-claim cluster in the covariate portfolio contains only 14 claims, those without legal representation, which fails the credibility criterion even though it is stable.

5.5 Experiment 3: other uses of groups, across the five scenarios

Table 6 and Figure 7 report the mean absolute RBNS error over $\tau = 24, \ldots, 40$ for each method and scenario.

Table 6: Mean absolute RBNS reserve error (%) over $\tau = 24, \ldots, 40$, by method and SPLICE scenario (standard portfolios). Groups are K-means clusters on notification features ($k = 3$), except the open-claim clusters ($k = 4$).

Method	C1	C2	C3	C4	C5	Mean
Raw case estimates	40.2	43.5	42.4	47.4	35.9	41.9
Paid chain ladder, unsegmented	15.0	18.6	21.7	24.5	26.7	21.3
Paid chain ladder by clusters	16.9	19.4	21.1	24.1	28.8	22.1
Paid chain ladder by random groups	15.1	18.8	21.7	24.0	27.2	21.4
Case-estimate factor, single	9.3	10.2	8.6	12.3	7.4	9.6
Case-estimate factors, bands	8.5	10.7	6.8	15.5	3.5	9.0
Case-estimate factors, open-claim clusters	8.7	10.6	8.2	14.2	5.2	9.4
Large claims apart, cluster	10.6	8.8	10.7	9.4	14.3	10.8
Large claims apart, rule	9.2	11.2	10.2	9.1	13.4	10.6
GLM, no trend	15.6	18.2	17.7	37.6	31.0	24.0
GLM, calendar trend	17.4	19.4	20.2	29.7	21.1	21.6
GLM, calendar trend, by clusters	17.7	19.4	21.3	27.2	23.0	21.7
GLM, calendar and speed trends	20.0	25.3	25.5	33.9	14.5	23.8
GLM, calendar and speed trends, by clusters	19.6	23.8	24.4	29.1	12.5	21.9

Splitting the triangle fails in every scenario. The clustered chain ladder is never better than the random groups. This includes Complexities 1 and 2, where no calendar-period trend can be blamed. The negative result of Section 5.2 is therefore not a feature of Complexity 5: in these simulations, the paid development patterns of the clusters are not different enough for separate projection to pay for the loss of data.

Adjusting case estimates is the most reliable approach. A single factor gives 7–12% in every scenario, and case-estimate factors are the best approach in Complexities 1, 3 and 5. Grouping

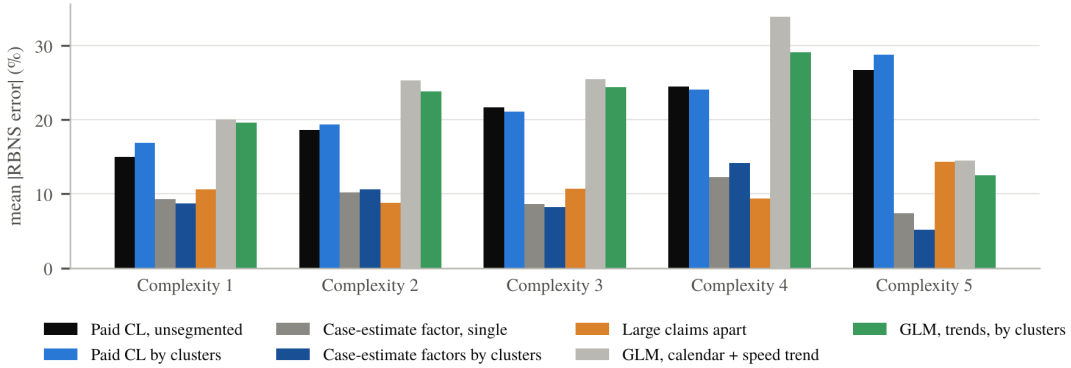


Figure 7: Mean absolute RBNS reserve error over $\tau = 24, \ldots, 40$ for the main methods, by SPLICE scenario.

does not improve it on average (9.4% against 9.6%), and simple bands do as well as clusters. The case estimates are short by 35–49% at $\tau = 40$ in every scenario, including Complexity 1, which has no inflation. The shortfall is therefore a property of how SPLICE generates case estimates, not an effect of inflation.

Treating large claims apart is the best approach when there is an inflation shock. In Complexities 2 and 4 it gives the lowest errors, 8.8% and 9.1–9.4%. In Complexity 4 that is far better than the paid chain ladder (24.5%) and the single case-estimate factor (12.3%). It reduces the chain-ladder error in every scenario, by between 29% (Complexity 1) and 62% (Complexity 4). A threshold rule on the relative initial estimate defines the large group as well as the cluster does.

Groups add value inside a model that handles the calendar effects. The calendar and speed trends help only where such trends exist. In Complexity 5 they reduce the GLM error from 31.0% to 14.5%. In Complexities 1–4 the speed trend extrapolates a pattern that is not there and increases the error. Within the model with both trends, however, group-specific development patterns reduce the error in every scenario (mean 21.9% against 23.8%; Complexity 4 29.1% against 33.9%; Complexity 5 12.5% against 14.5%). This is the one setting in which the clusters add value consistently: once the model accounts for the change over time, separating claim types refines it. The GLMs do not, however, beat the case-estimate factors in any scenario.

The groups make useful monitors (Figure 8).

- *No false alarm without change.* In Complexity 1 no cluster shows a significant trend in settlement speed (Spearman ρ 0.11–0.24, $p \geq 0.18$) or in its share of the portfolio.
- *A speed-up within the groups in Complexity 5.* Two of the three clusters settle significantly faster over time ($\rho = 0.51$, $p = 0.004$; $\rho = 0.42$, $p = 0.018$), and the share of the large cluster rises ($\rho = 0.60$, $p < 0.001$). The speed-up occurs within the groups, not only through the change in mix, which is why splitting the triangle cannot remove it.
- *Limited power with 30 cohorts.* In Complexity 3, where the speed-up is the designed feature, the signal is present in all three clusters but weak (ρ 0.26–0.35, p 0.06–0.16).

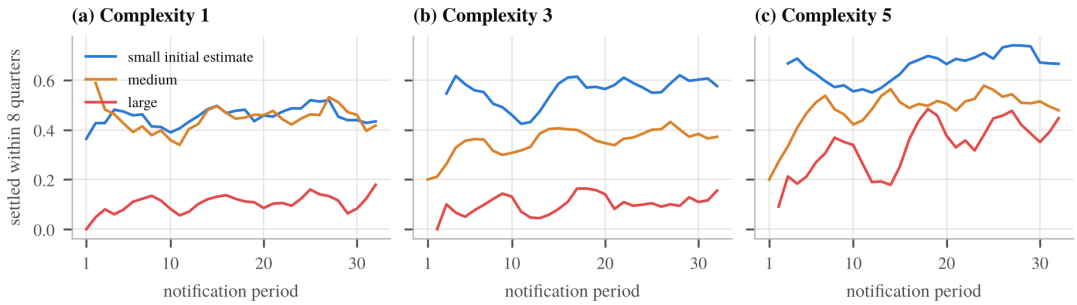


Figure 8: Monitoring by cluster: share of claims settled within eight quarters of notification, by notification cohort (four-period moving average), at $\tau = 40$.

6. Discussion and Conclusions

We asked whether clustering claims by their development characteristics yields segments that are useful to a reserving actuary. On the five SPLICE scenarios the answer depends on what the segments are used for. Clustering helps to understand a portfolio. It is not, by itself, a better way to compute a reserve.

6.1 Main findings

i. *The data contain one natural split.* It separates small, quickly settled claims from the rest. Clustering recovers it without supervision, and K-means does so much more stably than Ward. Finer partitions depend on how the correlated measures of claim complexity are weighted, which is a choice the analyst must justify.

ii. *Clusters do not improve a segmented chain ladder.* In none of the five scenarios is the clustered chain ladder better than random segmentations of the same sizes. This includes the scenarios without calendar-period trends. The clusters' development patterns are not different enough to pay for the loss of data in thinner triangles.

iii. *Adjusting case estimates is the most reliable approach.* Factors fitted out of time on the open claims give 7–12% error in every scenario, against 15–27% for the paid chain ladder. Most of the gain comes from a single factor. Groups add little on average, and no more than simple bands.

iv. *Groups help in three more specific roles.*
   - *Large claims apart.* Treating the large group with adjusted case estimates is the best approach when there is an inflation shock, although a threshold rule defines the group as well as the cluster does.
   - *Inside a model with calendar effects.* In a GLM with calendar and speed trends, group-specific development patterns reduce the error in every scenario. This is the only setting where the clusters add value consistently.
   - *Monitoring.* Tracking the groups over time distinguishes a change in mix from a change in development speed within the groups. It raised no false alarm in the scenario without change.

v. *Information timing, inflation and evaluation design matter as much as the algorithm.* In our first analysis, a cluster built on nominal amounts appeared to show a case-estimate shortfall of almost ten times; it disappeared once inflation drift was removed from the features, which also raised the stability of the clusters from 0.70 to 0.96. Covariate-based segments that beat random groups at $\tau = 40$ gave no gain over all valuation dates. A triangle segment must also be defined at a fixed development age, otherwise it confounds claim type with time since reporting.

6.2 Implications for practice

For an insurer, the evidence points to using claim groups to understand and manage the portfolio, rather than as a reserving method in their own right.

- *Prioritise case-estimate reviews.* Groups show where case estimates are least adequate. Here, newly reported claims with no payments yet ran off at about 2.6 times their case estimate, while old claims were estimated about right. Directing claims handlers to the weakest groups improves the input data before any reserving method is applied.
- *Monitor the portfolio.* A quarterly view of group shares and settlement speed answers a question that decides the method: has the mix of claims changed, or has the pace changed for everyone? The first calls for segmentation; the second for a method that models the trend.

- *Identify claims to reserve separately.* Clustering can reveal and justify a group of large, slow claims for separate treatment, such as the group of about a third of the claims that accounted for 69% of cost in Complexity 5. Once it is identified, a simple rule may define it as well.
- *Explain reserve movements.* Groups give a concrete vocabulary for explaining changes to management, auditors and regulators, for example "more large, slow claims were reported" or "case estimates on recent claims proved short".
- *Feed richer models.* Inside a model that accounts for calendar effects, the group is a useful variable. This suggests a role for clusters as inputs to individual-claim or neural-network models.

Whatever the use, the evaluation should be held to the five criteria of Section 2.4, which proved discriminating in practice:

- the stability test exposed Ward's fragility;
- the drift check exposed the nominal-amount artefact;
- the credibility check flagged a 14-claim cluster;
- the random-group placebo, applied over several valuation dates, showed that the apparent improvements of the segmented chain ladder were noise.

We recommend that any proposed clustering for reserving be reported against random segmentations of the same sizes and against the simplest rule-based alternative.

6.3 Limitations

- *Simulated data.* In SPLICE the case-estimate shortfall is large but very regular across claims and over time, which makes its correction unusually effective. In real portfolios case-estimate errors are likely to be less regular, and groups that capture differences in case-estimate practice could matter more.
- *Limited set of methods.* We used the paid chain ladder without a tail and a simple case-estimate factor that understates full development. We did not test incurred methods, Bornhuetter–Ferguson or individual-claim models. The GLM trends are linear and extrapolated, which helps where the simulated trend is linear and hurts where there is none.
- *No IBNR.* Claims not reported by the valuation date are excluded throughout.
- *Covariate timing.* The covariates are known from notification in the simulation, which may be optimistic for legal representation and injury severity in practice.
- *Low power of the monitor.* With 30 observable cohorts, the monitor detects a strong change in speed but only weakly a moderate one.
- *Software versions.* Simulated portfolios depend on the package versions: the development version of SynthETIC changes every draw after the covariate step (Appendix A). Data sets should be published, or their package versions documented, with the research.

6.4 Further work

The decisive next step is to apply the same framework to real data, where case-estimate practice and claim mix are less regular than in simulation. That framework consists of:

- valuation-date features;
- notification-period RBNS triangles;
- out-of-time evaluation;
- random-group placebos.

Other extensions are:

- combining case-estimate factors with a model of calendar effects;
- comparing against incurred methods;
- adding model-based clustering to the ex-post comparison;

- comparing the groups with the neural-network models used elsewhere in the working party, on the same valuation date.

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Appendix A Covariate data and reproducibility

The SPLICE repository publishes a covariate-adjusted version of each scenario, in which claim sizes depend on legal representation, injury severity and claimant age, but it does not publish the covariate assigned to each claim. The SynthETIC package distributes a covariate data set, `test_covariates_dataset`, but it labels SynthETIC's own example portfolio of 3,624 claims, whose per-period claim counts it matches exactly. It cannot be joined to the SPLICE Complexity 5 portfolio of 3,663 claims.

The SPLICE generator returns the covariate labels alongside the claims, so we regenerated the covariate portfolio with the published script (random seed 42) and exported the labels (repository script `scripts/export_splice_cov_labels.R`). We found the following, using R 4.3.3:

- *With the released versions* (SynthETIC 1.1.1 and SPLICE 1.1.2 from CRAN), the regenerated claim, payment and incurred tables match the published files to machine precision. This holds also when the generator is called twice in the same session. The labels we export therefore belong to the published files, and all covariate results in this paper use the published files with these labels.
- *With the development version of SynthETIC* (1.1.2 on GitHub, which changes how the random-number state is handled in the covariate step), the covariate labels and adjusted claim sizes are identical. Notification and settlement delays, payments and case-estimate histories, however, differ from the published files. An earlier draft of this work used that version by mistake.
- *For the same seed, the covariate and the standard portfolios differ* in everything drawn after the covariate step. This is expected, and the two should not be mixed.

Simulated data sets are thus reproducible for a given set of package versions, but not across them. We recommend publishing the data with the research, or documenting the versions used.

Table A.1: Published covariate portfolio: claims by injury severity.

Injury severity	Claims	Median claim size	Median settlement delay (quarters)
1	1,902	33,800	5.3
2	1,108	74,700	7.3
3	370	169,000	10.4
4	190	282,000	11.5
5	43	514,000	15.2
6	50	28,600	5.8

Claims without legal representation (97) occur only with severity 1 and are much smaller than average. The claim-size relativity decreases with claimant age.